TOR-AMPLITUDE BOUNDED BELOW IS EQUIVALENT TO BEING REPRESENTABLE BY A K-FLAT COMPLEX OF FLAT MODULES VANISHING BELOW THE BOUND

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Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file sites-cohomology.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The counted proof is the author's complete argument: the proof is written out in full: the choice of the K-flat resolution, the flatness of the truncation cokernel via the last-one-flat criterion, the K-flatness of the kernel and the truncated complex by two-out-of-three, and the pullback argument giving the uniformity stated in the last sentence.

Editorial note. The label lemma-bounded-below-tor-amplitude names the counted statement, and the proof environment immediately following it in the source is delivered verbatim as its proof.

Editorial note. The statement and proof are copied from the pinned chapter file; the five cited inputs are kept as external pointers with their canonical Stacks tags.

Editorial note. The preamble is the chapter's own retained minimal preamble (sources/stacks-sites-cohomology/retained-preamble.tex), a verbatim excerpt of the native preamble: the Xy-pic setup and the native theorem declarations with the mathematical macros.

Allowed standard prerequisites (editorial).

- K-flat complexes of modules on a ringed site and the two-out-of-three property. Same chapter, the notion of a K-flat complex, the existence of K-flat resolutions with flat terms, and the two-out-of-three lemma for short exact sequences of complexes. Supplies the representative K^bullet chosen at the start and the two applications of two-out-of-three showing L^bullet and E^bullet are K-flat.
- Flatness criterion on the last term of a bounded above complex. Same chapter, the lemma concluding flatness of the cokernel of a map of flat modules from the vanishing of negative tensor cohomology. Supplies flatness of N = Coker(K^{a-1} -> K^a) and of the image I in K^a.
- Pullback of a K-flat complex is K-flat. Same chapter, the lemma showing that the pullback of a K-flat complex of flat modules along a morphism of ringed sites is again K-flat with flat terms. Supplies the final assertion by pulling the whole construction back along an arbitrary morphism of ringed sites.

Source reference. Original lemma 06Y4: sources/stacks-sites-cohomology/pinned-original-source.tex, source lines 3207–3207. Prerequisite pointer invoked at proof line(s) 12489; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Date: stacks-project:sites-cohomology.tex:local-snapshot-20261011; GNU FDL 1.2 or later.

Source reference. Original lemma 0BYH: sources/stacks-sites-cohomology/pinned-original-source.tex, source lines 3097–3097. Prerequisite pointer invoked at proof line(s) 12521; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original lemma 06YD: sources/stacks-sites-cohomology/pinned-original-source.tex, source lines 3120–3120. Prerequisite pointer invoked at proof line(s) 12519; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original lemma 0653: sources/stacks-sites-cohomology/pinned-original-source.tex, source lines 12403–12403. Prerequisite pointer invoked at proof line(s) 12498; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original lemma 06YC: sources/stacks-sites-cohomology/pinned-original-source.tex, source lines 3508–3508. Prerequisite pointer invoked at proof line(s) 12526; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Lemma 0.1. *Let $(\mathcal{C}, \mathcal{O})$ be a ringed site. Let E be an object of $D(\mathcal{O})$. Let $a \in \mathbf{Z}$. The following are equivalent*

1. *E has tor-amplitude in $[a, \infty]$.*
2. *E can be represented by a K-flat complex $\mathcal{E}^\bullet$ of flat $\mathcal{O}$-modules with $\mathcal{E}^i = 0$ for $i \notin [a, \infty]$.*

Moreover, we can choose $\mathcal{E}^\bullet$ such that any pullback by a morphism of ringed sites is a K-flat complex with flat terms.

Proof. The implication (2) $\Rightarrow$ (1) is immediate. Assume (1) holds. First we choose a K-flat complex $\mathcal{K}^\bullet$ with flat terms representing E, see Lemma 06Y4. For any $\mathcal{O}$-module $\mathcal{M}$ the cohomology of

$$\mathcal{K}^{n-1} \otimes_{\mathcal{O}} \mathcal{M} \to \mathcal{K}^n \otimes_{\mathcal{O}} \mathcal{M} \to \mathcal{K}^{n+1} \otimes_{\mathcal{O}} \mathcal{M}$$

computes $H^n(E \otimes^{\mathbf{L}}_{\mathcal{O}} \mathcal{M})$. This is always zero for $n < a$. Hence if we apply Lemma 0653 to the complex $\ldots \to \mathcal{K}^{a-1} \to \mathcal{K}^a \to \mathcal{K}^{a+1}$ we conclude that $\mathcal{N} = \mathrm{Coker}(\mathcal{K}^{a-1} \to \mathcal{K}^a)$ is a flat $\mathcal{O}$-module. We set

$$\mathcal{E}^\bullet = \tau_{\geq a}\mathcal{K}^\bullet = (\ldots \to 0 \to \mathcal{N} \to \mathcal{K}^{a+1} \to \ldots)$$

The kernel $\mathcal{L}^\bullet$ of $\mathcal{K}^\bullet \to \mathcal{E}^\bullet$ is the complex

$$\mathcal{L}^\bullet = (\ldots \to \mathcal{K}^{a-1} \to \mathcal{I} \to 0 \to \ldots)$$

where $\mathcal{I} \subset \mathcal{K}^a$ is the image of $\mathcal{K}^{a-1} \to \mathcal{K}^a$. Since we have the short exact sequence $0 \to \mathcal{I} \to \mathcal{K}^a \to \mathcal{N} \to 0$ we see that $\mathcal{I}$ is a flat $\mathcal{O}$-module. Thus $\mathcal{L}^\bullet$ is a bounded above complex of flat modules, hence K-flat by Lemma 06YD. It follows that $\mathcal{E}^\bullet$ is K-flat by Lemma 0BYH.

Proof of the final assertion. Let $f : (\mathcal{C}', \mathcal{O}') \to (\mathcal{C}, \mathcal{O})$ be a morphism of ringed sites. By Lemma 06YC the complex $f^*\mathcal{K}^\bullet$ is K-flat with flat terms. The complex $f^*\mathcal{L}^\bullet$ is K-flat as it is a bounded above complex of flat $\mathcal{O}'$-modules. We have a short exact sequence of complexes of $\mathcal{O}'$-modules

$$0 \to f^*\mathcal{L}^\bullet \to f^*\mathcal{K}^\bullet \to f^*\mathcal{E}^\bullet \to 0$$

because the short exact sequence $0 \to \mathcal{I} \to \mathcal{K}^a \to \mathcal{N} \to 0$ of flat modules pulls back to a short exact sequence. By Lemma 0BYH. the complex $f^*\mathcal{E}^\bullet$ is K-flat and the proof is complete. $\square$